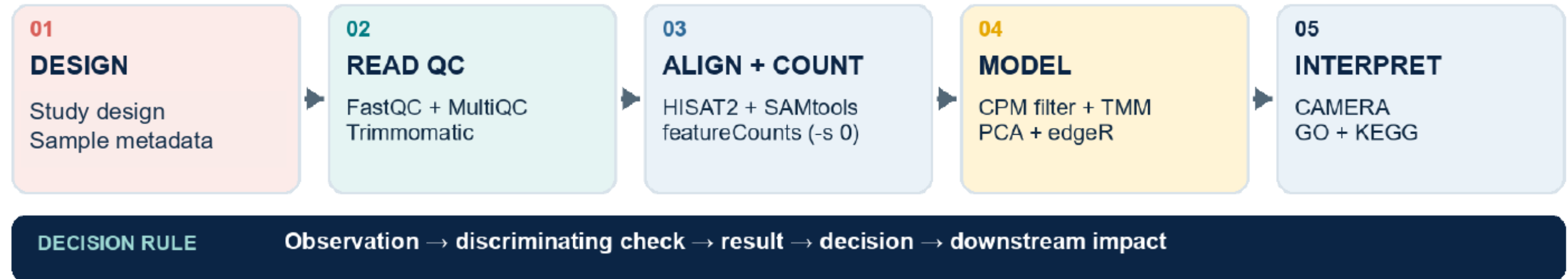


A reproducible RNA-seq analysis workflow

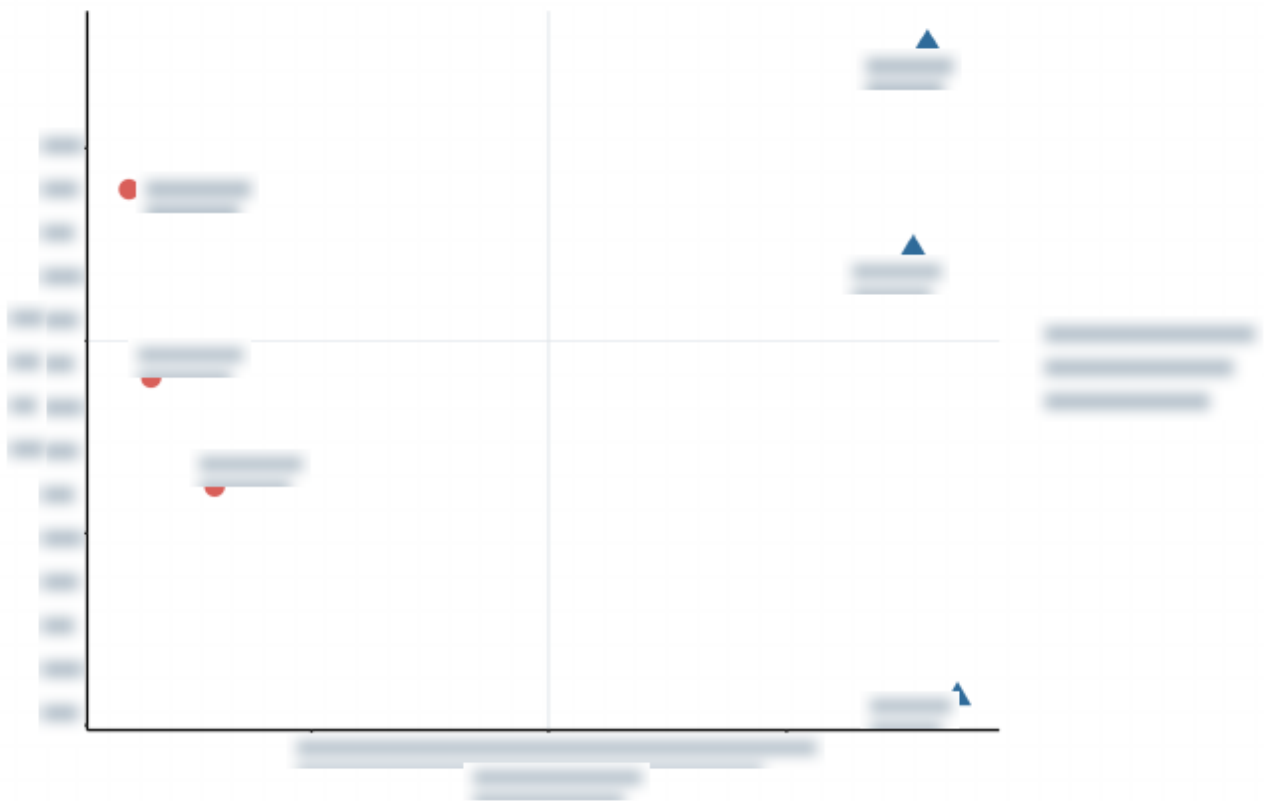
Quality control, alignment, statistical modelling and visual reporting.

From raw reads to interpretation

A documented workflow with explicit analysis decisions



Exploratory sample structure



PRINCIPAL COMPONENT ANALYSIS

Explore relationships among samples using PCA.
Labels, variance estimates and the biological conclusion
are withheld in this portfolio preview.

SKILLS DEMONSTRATED

Connect experimental design to quality control.
Assess sample structure before statistical modelling.
Explain assumptions and limits alongside the figures.

TOOLS ARE TIED TO PURPOSE

FastQC/MultiQC screened raw and paired-output reads; Trimmomatic applied quality trimming with MINLEN:50 and no ILLUMINACLIP (Andrews, n.d.; Bolger et al., 2014; Ewels et al., 2016). HISAT2/GRCm39, SAMtools and paired-fragment featureCounts (-s 0) supported alignment and counting (Danecek et al., 2021; Kim et al., 2019; Liao et al., 2014). TMM/edgeR modelled expression; CAMERA/limma and GO/KEGG supplied gene-set views (Robinson et al., 2010; Robinson & Oshlack, 2010; Wu & Smyth, 2012).

Visualising differential expression

Volcano plot design, model interpretation and responsible reporting.

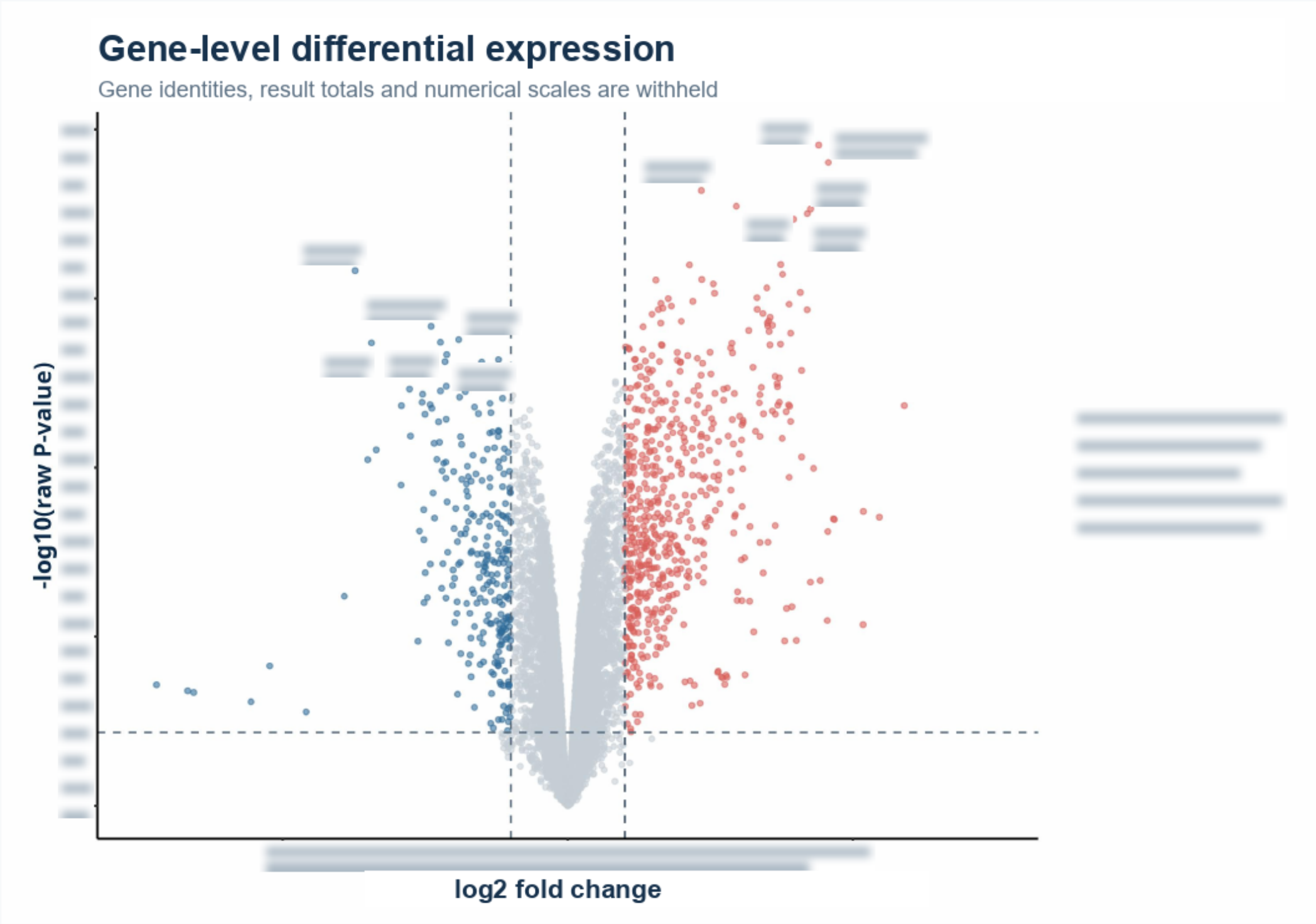


FIGURE 2 Gene-level change across the stated contrast

Portfolio preview of an original volcano plot. Gene names, selected-gene counts and biological interpretation are withheld. Chart marks remain visible to demonstrate the visual analysis.

Gene-set analysis and interpretation

Competitive testing and over-representation analysis provide complementary views.

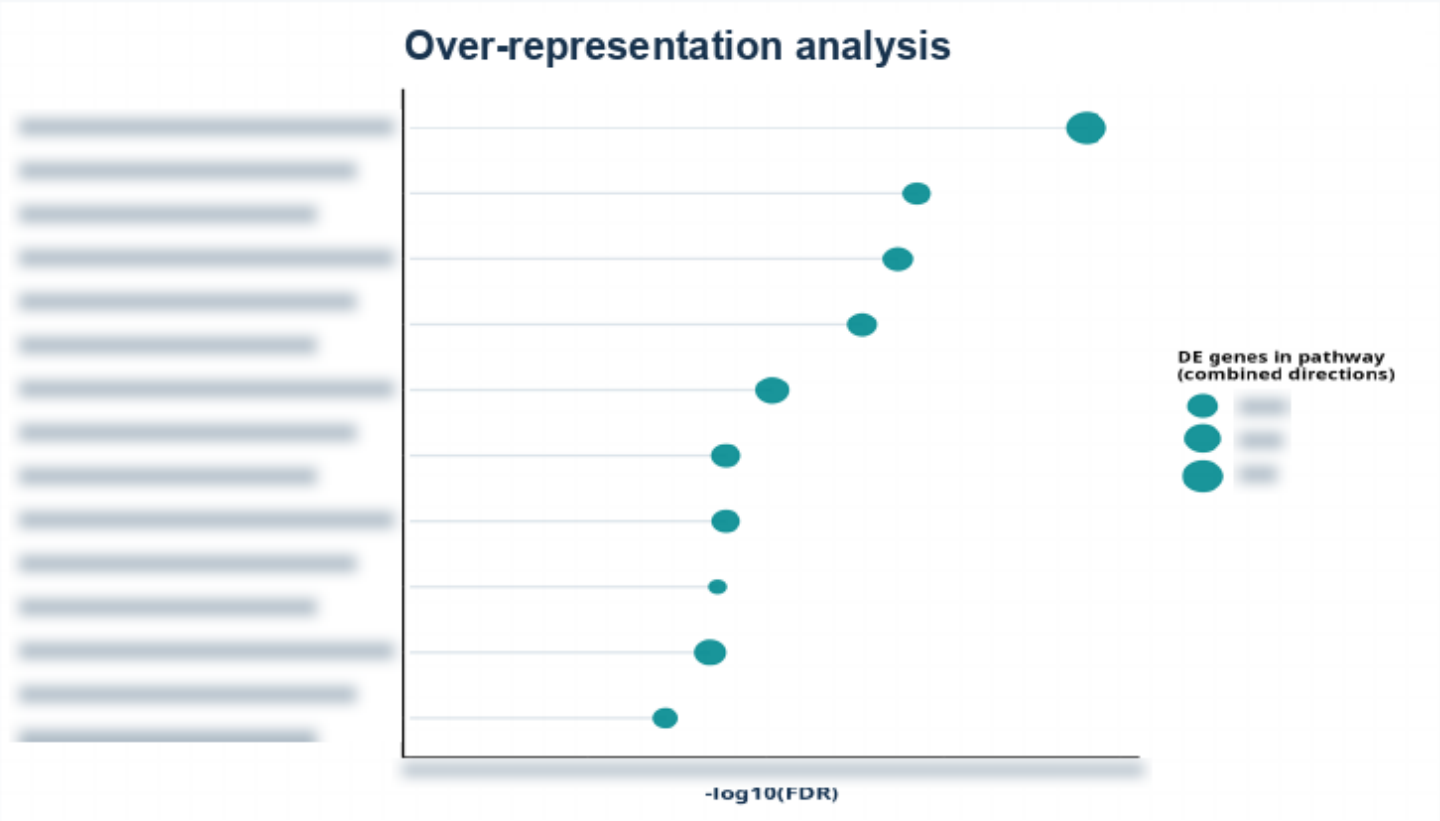
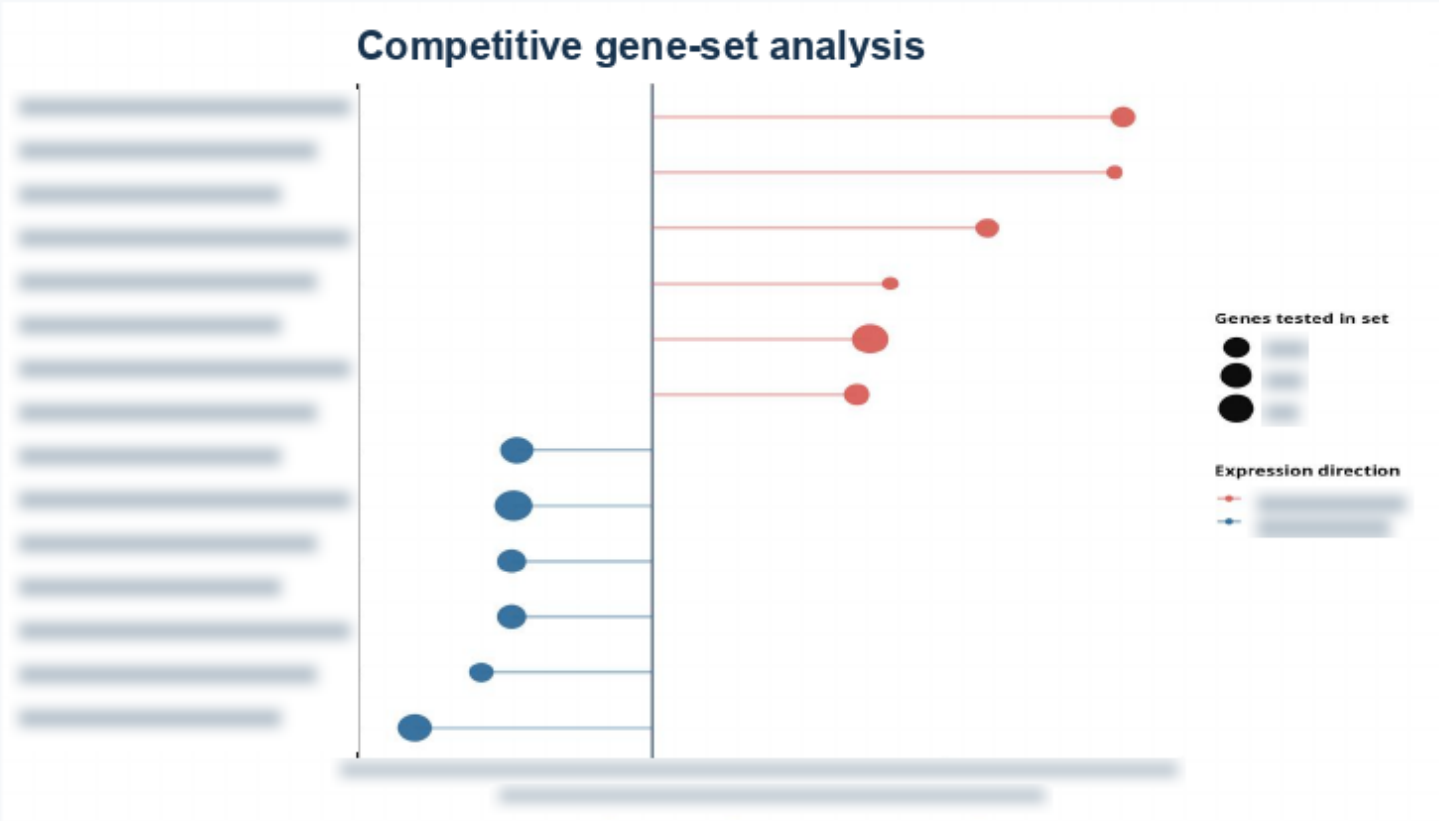


FIGURE 3 Direction-aware competitive enrichment

CAMERA evaluates gene sets while accounting for correlation among genes. Pathway identities, numerical results and condition labels are withheld.

BIOLOGICAL INTERPRETATION WITHHELD

RESULTS WITHHELD

FIGURE 4 Pooled-direction supporting context

Over-representation analysis complements the gene-set view. Pathway identities, gene counts and enrichment results are withheld in this portfolio preview.

CONCLUSIONS WITHHELD

APA REFERENCES

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SOFTWARE FastQC 0.12.1 · MultiQC 1.23 · Trimmomatic 0.38 · HISAT2 2.1.0 · samtools 1.19.2 · featureCounts 2.1.1 · R 4.4.3 · edgeR 4.4.2 · limma 3.62.2 · ggplot2 3.5.2 · org.Mm.eg.db 3.20.0

Read and library quality-control review

Diagnose technical issues and explain how they affect downstream analysis.

Read-quality profile

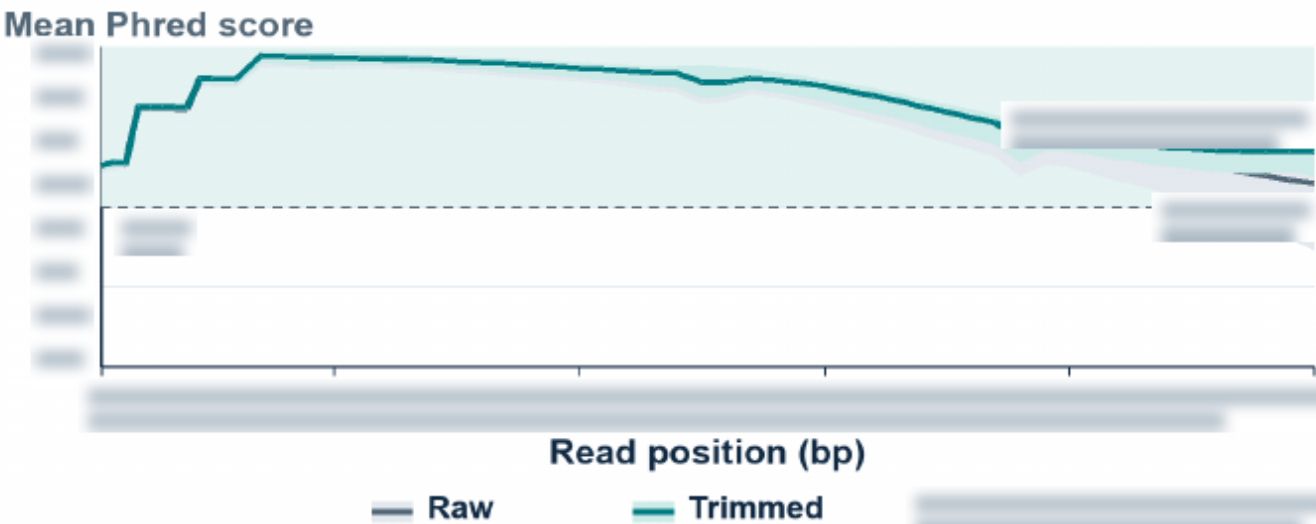


FIGURE 5 Read-quality assessment

Compare read-quality profiles before and after processing. Numerical measurements and the QC decision are withheld.

Nucleotide-composition profile

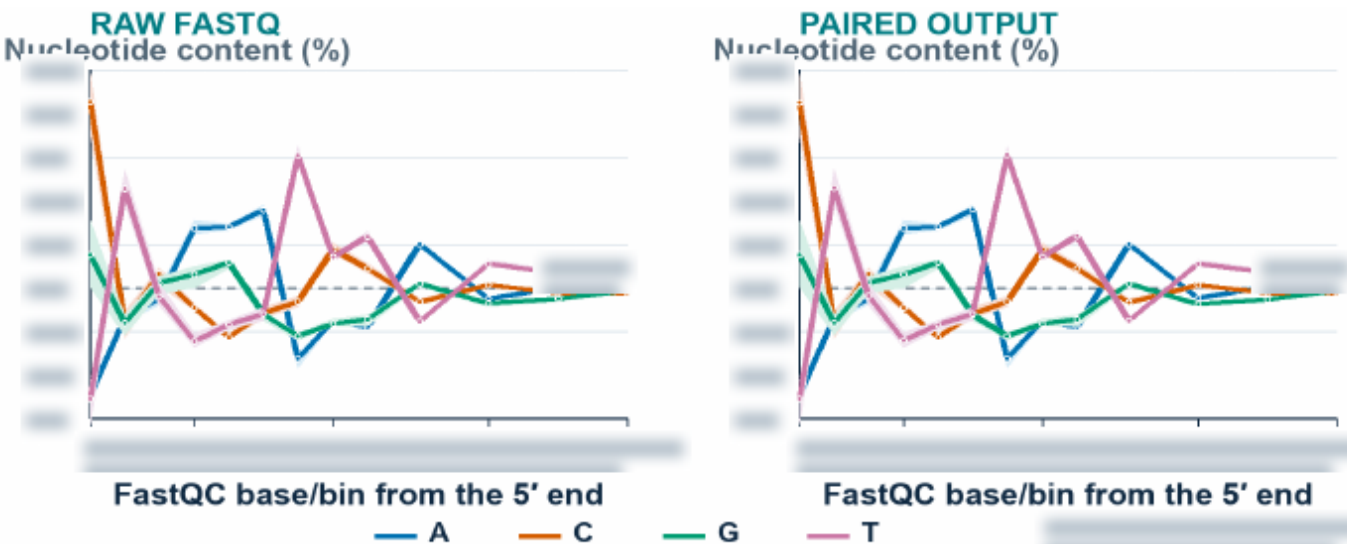


FIGURE 6 Composition assessment

Inspect nucleotide composition across the read. Numerical measurements and interpretation are withheld.

Paired-read length distribution



FIGURE 7 Read-length assessment

Review read-length distributions and mate consistency. Sample identities and quantitative findings are withheld.

Library complexity diagnostics

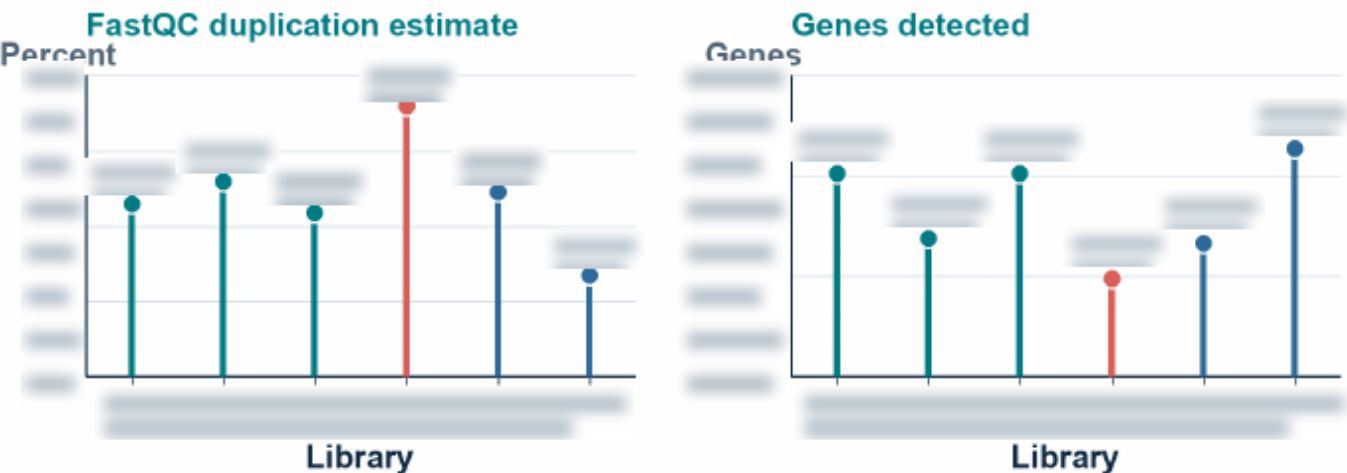


FIGURE 8 Complexity assessment

Assess duplication alongside expression breadth. Sample identities and quantitative findings are withheld.